

MRI Full Course Formula Sheet

Compact formulas for exam memorization - two lectures per page

Week 1 - NMR Basics

Spin		
Angular momentum	$\mathbf{p} = m\mathbf{v}, \mathbf{J} = \mathbf{r} \times \mathbf{p}$	Rotating charge.
Magnetic moment	$\boldsymbol{\mu} = \gamma \mathbf{J}$	γ gyromagnetic ratio.
Quantization	$J_z = \hbar m, m = l - 1, \dots, -l$	Spin projection.
Spin 1/2	$l = 1/2 \Rightarrow m = \pm 1/2$	^1H has two states.
Moment projection	$\mu_z = \gamma \hbar m = \pm (1/2) \gamma \hbar$	Along B_0 .

Energy And Resonance		
Energy	$E_m = -\mu_z B_0 = -\gamma \hbar B_0 m$	Parallel lower for $\gamma > 0$.
Gap	$\Delta E = \gamma \hbar B_0 = \hbar \omega_0$	Energy splitting.
Boltzmann	$n_{\text{down}}/n_{\text{up}} = \exp(-\Delta E/k_B T)$	Population ratio.
Polarization	$\Delta n/n \approx \Delta E/(2k_B T)$	For $\Delta E \ll k_B T$.
Net M	$M_0 = \sum \mu = \Delta n \mu_z e_z$	Longitudinal M.
Scaling	$M_0 \propto \rho_H B_0 / T$	Proton density matters.

Dynamics		
Torque	$d\boldsymbol{\mu}/dt = \boldsymbol{\gamma}(\boldsymbol{\mu} \times \mathbf{B})$	Precession.
Macroscopic	$d\mathbf{M}/dt = \boldsymbol{\gamma}(\mathbf{M} \times \mathbf{B})$	Same law for M.
RF	$\mathbf{B}_1(t) = B_1 \cos(\omega_0 t) \mathbf{e}_x = B_L + B_R$	Excitation field.
Rotating frame	$(d\mathbf{M}/dt)_{\text{rot}} = \boldsymbol{\gamma} \mathbf{M} \times \mathbf{B} - \omega_0 \times \mathbf{M} = \boldsymbol{\gamma} \mathbf{M} \times \mathbf{B}_{\text{eff}}$	Use effective field.
Effective field	$\mathbf{B}_{\text{eff}} = (B_0 - \omega/\gamma) \mathbf{e}_z + B_{1,\text{rot}} \mathbf{e}_x$	On resonance mostly B1.
Flip angle	$\alpha = \gamma \int B_{1,\text{rot}}(t) dt$	RF pulse area.

Relaxation		
T2*	$1/T_2^* = 1/T_2 + \gamma \delta B$	Field inhomogeneity.
Bloch x	$dM_x/dt = \gamma(M_y B_z - M_z B_y) - M_x/T_2$	Transverse loss.
Bloch y	$dM_y/dt = \gamma(M_z B_x - M_x B_z) - M_y/T_2$	Transverse loss.
Bloch z	$dM_z/dt = \gamma(M_x B_y - M_y B_x) - (M_z - M_0)/T_1$	Recovery.

Week 2 - Image Formation

Encoding		
Gradient	$G_i = \partial B_z / \partial i$	$i = x, y, z$.
Loop field	$B_z(z) = \mu_0 I r^2 / (2(r^2 + z^2)^{3/2})$	Circular current loop.
Linear field	$\Delta B_z(x) = G_x x$	Constant slope.
Frequency	$\omega(r) = \gamma(B_0 + G \cdot r)$	Position dependent.
Readout	$\Delta\omega(x) = \gamma G_x x$	Frequency encoding.
Phase	$\phi(y) = \gamma G_y y T_y$	Phase encoding.
K-space	$\mathbf{k}(t) = \gamma \int \mathbf{G}(t) dt$	Constant G: $\mathbf{k} = \gamma \mathbf{G} t$.

Signal And FT		
Signal	$s(k_x, k_y) \propto \sum_{i,j} \rho(x_i, y_j) \exp[i(k_x x_i + k_y y_j)]$	FT of object.
Image	$\rho(x, y) \propto \sum_{p,q} s(k_{x,p}, k_{y,q}) \exp[-j(k_{x,p} x + k_{y,q} y)]$	Inverse FT.
K spacing	$\Delta k = 2\pi / \text{FOV} = \text{BW} / N_x$	Nyquist/readout spacing.
Pixel	$\Delta x = \text{FOV}_x / N_x$	Same in y.
Max k	$k_{\text{max}} = \pi / \Delta x = \pi N_x / \text{FOV}_x$	Resolution.
Readout BW	$\text{BW}_x = \gamma G_x \text{FOV}_x$	Angular freq convention.
Dwell	$\Delta t = 2\pi / \text{BW}_x, T_x = N_x \Delta t$	Sampling.

Slice And Contrast		
Gradient moment	$k_i = \gamma \int G_i(t) dt$	Gradient area.
Slice thickness	$\Delta z = \text{BW}_{\text{RF}} / (\gamma G_z)$	RF bandwidth.
RF selectivity	$R = \text{BW}_{\text{RF}} T_{\text{RF}}$	Longer pulse narrower BW.
PD contrast	$C \propto \rho_A - \rho_B$	Long TR.
Partial sat.	$C \propto \rho_A [1 - \exp(-\text{TR}/T_{1A})] - \rho_B [1 - \exp(-\text{TR}/T_{1B})]$	T1 weighting.
Scan time	$T_{\text{scan}} = N_{\text{phase}} \text{TR}$	One ky line per TR.

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Week 3 - Fast And Parallel Imaging

Speed, SNR, Resolution		
Voxel	$\Delta V = \Delta x \Delta y \Delta z$	Voxel volume.
Signal	$u_s(t) \approx \omega_0 \mu_0 \rho \gamma^2 \hbar^2 B_0 / (4k_B T)$	Induced voltage.
Noise std.	$\psi_n \approx \gamma B_0$	Receive noise scale.
SNR physical	$SNR = u_s / \psi_n \approx \gamma^2 B_0 \eta \approx \Delta V \sqrt{t}$	Field, coil, voxel, time.
SNR scaling	$SNR \propto \Delta V \sqrt{T_{scan}} / \sqrt{BW}$	Core tradeoff.
Averages	$SNR \propto \sqrt{NSA}$	Repeated scans.
Acceleration	$T_{scan,R} \approx T_{scan} / R$	Undersampling.
Aliased FOV	$FOV_{alias} = FOV / R$	Phase undersampling.

Coil Encoding

Coil signal	$s_v(k) = \int \rho(x) c_v(x) \exp(ikx) dx$	Sensitivity cy.
Coil sens.	$C(r) = B_1(r), B(r) = \mu_0 I / (4\pi) \int d\mathbf{r}' \alpha(r-r') / r-r' ^3$	Biot-Savart law.
Matrix model	$\mathbf{s} = \mathbf{E} \rho + \eta$	Encoding plus noise.
Decode	$\mathbf{i} = \mathbf{F} \mathbf{s}, \mathbf{F} \mathbf{E} = \mathbf{I}$ (ideal)	E is not directly invertible.
Pseudoinverse	$\mathbf{F} = (\mathbf{E}^H \mathbf{E})^{-1} \mathbf{E}^H$	Noise ignored.
SENSE	$\hat{\rho} = (\mathbf{E}^H \Psi^{-1} \mathbf{E})^{-1} \mathbf{E}^H \Psi^{-1} \mathbf{s}$	Accounts for non-independent coil noise.
Regularized	$\hat{\rho} = (\mathbf{E}^H \Psi^{-1} \mathbf{E} + \lambda \mathbf{I})^{-1} \mathbf{E}^H \Psi^{-1} \mathbf{s}$	Stabilizes ill-conditioning.

Parallel Limits

SENSE SNR	$SNR_{SENSE} = SNR_{full} (\sqrt{R} \cdot g(x))$	g-factor penalty.
g-factor	$g(x) = \sqrt{[(\mathbf{E}^H \mathbf{E})_{ii} ((\mathbf{E}^H \mathbf{E})^{-1})_{ii}] \geq 1}$	Noise amplification.
Good coils	Distinct $c_v(x) \Rightarrow$ low g	Separates aliased voxels.
Failure	$R > N_{coils}$ or poor sensitivities	Underdetermined/ill-conditioned.

Week 4 - Image Contrast

Relaxation Contrast		
T1 recovery	$M_z(t) = M_0 + [M_z(0) - M_0] \exp(-t/T_1)$	Longitudinal.
After 90°	$M_z(t) = M_0 [1 - \exp(-t/T_1)]$	$M_z(0) = 0$.
T2 decay	$M_{xy}(t) = M_{xy}(0) \exp(-t/T_2)$	Spin-spin.
T2* decay	$M_{xy}(t) = M_{xy}(0) \exp(-t/T_2^*)$	GRE/FID.
Dipolar T1	$1/T_1 = [6\hbar^2 \gamma^4 / (20\sigma^6)] [J(\omega) + 4J(2\omega)]$	Two spins at distance r.
Dipolar T2	$1/T_2 = [3\hbar^2 \gamma^4 / (20\sigma^6)] [3J(0) + 5J(\omega) + 2J(2\omega)]$	Includes static term J(0).

Analytical Description

Sequence ops	$M_{n+1} = O_n \dots O_1 M_0, O \in \{A(\alpha), R(\varphi)\}$	RF pulse plus relaxation blocks.
RF operator	$A(\alpha) = \cos \alpha \mathbf{I} + (1 - \cos \alpha) b b^T + \sin \alpha \text{skew}(b)$	Rotation around RF axis b.
Relax. op.	$M' = R(\varphi) M + (1 - E_1) M_0 e_z$	$E_1 = e^{-t/T_1}, E_2 = e^{-t/T_2}$.
R matrix	$R(\varphi) = [E_2 \cos \varphi, -E_2 \sin \varphi, 0; E_2 \sin \varphi, E_2 \cos \varphi, 0; 0, 0, E_1]$	Precession plus decay.

Common Signals

Spin echo	$S_{SE} \approx \rho [1 - \exp(-TR/T_1)] \exp(-TE/T_2)$	T1 by TR, T2 by TE.
Max contrast TR	$T_R = [T_{1A} T_{1B} / (T_{1B} - T_{1A})] \ln[(S_A T_{1B}) / (S_B T_{1A})]$	Obtained by setting dC/dTR=0.
GRE	$S_{GRE} \approx \rho \sin \alpha (1 - E_1) / (1 - E_1 \cos \alpha) \exp(-TE/T_2^*)$	$E1 = \exp(-TR/T1)$.
Inversion	$M_z(TI) = M_0 [1 - 2 \exp(-TI/T_1)]$	TR >> T1.
Null time	$T_{null} = T_1 \ln 2$	Suppress tissue.
Ernst	$\alpha_E = \arccos[\exp(-TR/T_1)]$	Max steady-state signal.

Weighting Rules

PD	TR long, TE short	Minimize relaxation weighting.
T1	TR short, TE short	T1 recovery dominates.
T2	TR long, TE long	T2 decay dominates.
T2*	GRE with TE long	Sensitive to dephasing.

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Week 5 - SNR And Hardware

Signal And Noise		
Signal voltage	$U_{sig}(x)=\omega M_{xy}(x)C(x)\Delta V$	Receive sensitivity C.
Noise variance	$\Psi_U=4k_B BW(T_s R_s + T_c R_c)$	U_{noise} variance; coil term small except tiny coils.
SNR	$SNR=U_{signal}/\sqrt{\Psi_U}=\omega M_{xy} C \Delta V / \sqrt{N_{avg} [4k_B BW(T_s R_s + T_c R_c)]}$	Full voltage-noise form.
SNR scaling	$SNR \propto \Delta V \sqrt{NSA} / \sqrt{BW}$	Resolution/speed tradeoff.

Optimization		
Increase signal	$\uparrow C, \uparrow M_{xy}, \uparrow \omega, \uparrow \Delta V$	More voltage.
Reduce noise	$\downarrow BW, \downarrow T, \downarrow R_{coil}$	Less thermal noise.
Field scaling	$M_0 \propto B_0, \omega \propto B_0$	U_{sig} roughly grows strongly with B_0 .
Surface coil	$C(x)$ high near coil	High local SNR.

Resolution Limits		
Fourier	$\Delta x \approx \pi / k_{max}$	Sampling aperture.
Readout filter	$H(k)=rect[k/(2k_{max})] \exp[-(k+k_{max})/k^*]$	Relaxation-weighted k-space.
Relaxation blur	$k^* = \gamma G T_2^*$	Finite T_2^* filters k-space.
Diffusion blur	$x^2 \approx 6 D T_{acq}$	Long readouts blur.
Relax. limit	$S(T_2^*) \approx \exp(-t/T_2^*)$	Resolution improves $\propto G^{-1}$.
Diff. limit	$S(D) \approx \exp(-\gamma^2 G^2 D t^3)$	Resolution improves $\propto G^{-1/3}$.
SoS combine	$I_{SoS} = \sqrt{(\sum_c I_c)^2}$	Magnitude coil combine.

Week 6 - Flow Imaging

Motion Phase		
Trajectory	$r(t)=r_0+v(t-t_0)+a(t-t_0)^2/2+\dots$	Moving spin.
Phase	$\phi=\gamma \int G(t)r(t)dt$	Gradient phase.
Moments	$M_n=\int G(t)(t-t_0)^n dt$	nth gradient moment.
Expansion	$\phi=\gamma[r_0 M_0 + v M_1 + a M_2/2 + \dots]$	Position/velocity/acceleration.
Bipolar	$M_0=0, \phi=\gamma v M_1$	Velocity encoding.

Amplitude Contrast		
GRE/PS	$\sigma_{PS} \approx M_0 [v_0 T_R + (\Delta z - v_0 T_R)(1 - e^{-T_R/T_1})]$	Inflow plus partial saturation.
Spin echo	$\sigma_{SE} = \sigma_{PS} e^{-T_E/T_2} (1 - T_E v_0 / (2 \Delta z))$	T_2 decay; spins must stay in slice for 90/180/readout.

Phase Contrast		
Encoding	$VENC = \pi / (\gamma M_1)$	Phase reaches $\pm \pi$.
Velocity	$v = (\Delta \phi / \pi) VENC$	Phase difference map.
Aliasing	$ v > VENC \Rightarrow$ phase wraps	Set VENC high enough.
Flow rate	$Q = \sum_i v_i \Delta A_i$	Through-plane flow.
Velocity distribution	$s(x, k_v) = \sum p(x, v) \exp(j k_v v)$	Generalized velocity encoding.

Hemodynamics		
Poiseuille	$Q = \Delta P / R, R \propto 1/d^4$	Diameter dominates.
Reynolds	$Re = \rho v d / \mu$	Laminar if $Re < \sim 2000$.
Shear stress	$SS = -\mu \partial v / \partial r$	Wall shear.
Contrast agent	$1/T_{1,app} = 1/T_1 + R_1 c$	Relaxivity model.

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Week 7 - Motion And Artifacts

Motion Encoding		
Phase	$\varphi = \gamma \int G(t) x(t) dt$	All motion artifacts start here.
Motion model	$x(t) = x_0 + v(t-t_0) + a(t-t_0)^2/2 + \dots$	Taylor expansion.
Moment nulling	$M_0 = 0$ removes position phase; $M_1 = 0$ compensates velocity	Gradient design.
Velocity phase	$\varphi_v = \gamma v M_1$	Residual if M_1 nonzero.
Velocity spread	signal $\propto \text{sinc}(\beta \gamma \Delta t \Delta y/2)$	Intravoxel dephasing.

Artifact Rules		
Intra-TR	motion during readout/echo $\Rightarrow$ phase errors and signal loss	Flow, pulsation.
Inter-TR	motion between ky lines $\Rightarrow$ ghosting/blurring	Respiration/patient.
Periodic ghosts	ghost spacing in phase direction $\propto 1/(f_{\text{motion}} \text{TR})$	Regular motion.
Random motion	random ky phase $\Rightarrow$ diffuse blur	Irregular motion.

Correction		
Rigid transform	$R(r,t) = A(t)r + d(t)$	Rotation plus translation.
Prospective gradients	$G'(t) = A(t)^{-T} G(t)$	Follow anatomy.
Phase correction	$\varphi'(t) = \varphi(t) - \gamma G(t) \cdot d(t)$	Translation compensation.

Week 8 - fMRI And Diffusion

BOLD/fMRI		
T2*	$1/T_2^* = 1/T_2 + \gamma \Delta B$	Static dephasing.
Susceptibility	$\Delta B \approx \Delta \chi B_0$	Blood oxygenation effect.
GRE BOLD	$S(TE) = S_0 \exp(-TE/T_2^*)$	TE near T2*.
Small change	$\Delta S/S \approx -TE \cdot \Delta R_2^*$	$R2^* = 1/T2^*$.

Diffusion Physics		
Fick	$j = -D \nabla c(r,t)$	Diffusion flux.
Diffusion eq.	$\partial c / \partial t = D \nabla^2 c$	Conservation plus Fick.
RMS displacement	$R_{\text{rms}} = \sqrt{6D\Delta t}$	3D free diffusion.
Einstein	$D = \sqrt{2} l^2 / 6$	Random walk.

DWI/DTI		
DWI signal	$S(TE,b) = S_0 \exp(-TE/T_2^*) \exp(-bD)$	Scalar diffusion.
b-value	$b = \gamma^2 G^2 \delta^2 (\Delta - \delta/3)$	PGSE sensitivity.
Tensor	$\ln(S/S_0) = -b \ g^T D g$	Direction g.
MD	$MD = (\lambda_1 + \lambda_2 + \lambda_3)/3$	Mean diffusivity.
FA	$FA = \sqrt{(3/2) \left(\sqrt{\sum (\lambda_i - MD)^2} / \sqrt{\sum \lambda_i^2} \right)}$	Anisotropy.

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Week 9 - Advanced Imaging

General Encoding		
Data model	$\mathbf{d} = \mathbf{E}\mathbf{p} + \boldsymbol{\eta}$	General MRI inverse problem.
Cartesian FT	$\mathbf{E} = \text{Fourier sampling operator}$	Standard MRI.
Parallel	$\mathbf{E}$ includes coil sensitivities $c_v(r)$	SENSE/arrays.
Optimal inverse	$\hat{\mathbf{p}} = (\mathbf{E}^H \boldsymbol{\Psi}^{-1} \mathbf{E})^{-1} \mathbf{E}^H \boldsymbol{\Psi}^{-1} \mathbf{d}$	If well-conditioned.
Regularized	$\hat{\mathbf{p}} = (\mathbf{E}^H \boldsymbol{\Psi}^{-1} \mathbf{E} + \lambda \mathbf{I})^{-1} \mathbf{E}^H \boldsymbol{\Psi}^{-1} \mathbf{d}$	For high R.

Compressed Sensing		
Sparsity	$\mathbf{x} = \Phi \mathbf{p}$ has many small/zero coefficients	Sparse transform.
Undersampled data	$\mathbf{d}_0 = \mathbf{P}_0 \Phi \mathbf{p}$	Sample subset Ω .
CS recon	$\min_{\mathbf{p}} \ \Phi \mathbf{p}\ _1 \text{ s.t. } \ \mathbf{E} \mathbf{p} - \mathbf{d}\ _2 \leq \epsilon$	Sparse recovery.
TV	$\Phi \mathbf{p} = \nabla \mathbf{p} \Rightarrow$ total variation penalty	Piecewise smooth images.
Sampling	random/incoherent k-space undersampling	Artifacts become noise-like.

Limits		
PSF	$\text{PSF}(\mathbf{r}) = \mathbf{F} \mathbf{T}^{-1} \{\text{sampling pattern}\}$	Undersampling artifact shape.
Acceleration	higher R needs sparsity + SNR + calibration	No free lunch.
Low-rank/dynamics	suppp small or temporal basis small	Advanced priors.

Week 10 - Spectroscopy I

Chemical Shift		
Larmor	$\nu = -\gamma B_0$ [Hz]	Sign by convention.
Shift	$\delta = (\nu - \nu_{\text{ref}}) / \nu_{\text{ref}} \cdot 10^6 \text{ ppm}$	Field-independent ppm.
Hz separation	$\Delta \nu = \Delta \delta \cdot \nu_{\text{ref}} \cdot 10^{-6}$	Grows with B_0 .
1 ppm	1 ppm = $\nu_{\text{ref}} \cdot 10^{-6} \text{ Hz}$	64 Hz at 1.5T for ^1H .

Spectrum		
FID	$s(t) = A \exp(-t/T_2^*) \exp(i2\pi \nu_A t)$	Single component.
Spectrum	$S(\nu) = \text{FT}\{s(t)\}$	Frequency-domain signal.
Lorentzian	$\text{Re } S(\nu) \propto T_2^* / [1 + (2\pi T_2^* (\nu - \nu_A))^2]$	Line shape.
Linewidth	$\text{FWHM} = 1/(\pi T_2^*)$	Shorter T_2^* broader.
Mixture	$s(t) = \sum_c w_c \exp(-t/T_{2,c}^*) \exp(i2\pi \nu_c t)$	Superposition.

Sampling And J		
Acquisition	$T_{\text{acq}} = N \text{dt}$	N samples.
Bandwidth	$\text{BW} = 1/\text{dt}$	Hz.
Resolution	$d\nu = 1/T_{\text{acq}}$	Peak spacing.
Resolve peaks	$d\nu \leq \Delta \nu/2$	Rule of thumb.
J multiplet	N coupled spin-1 nuclei $\Rightarrow 2NI + 1$ lines	Binomial intensities.
J echo	$\text{TE} = n/J$	Odd n flips doublet sign.
Editing	$\text{TE} = 1/J$	Difference editing.

Week 11 - Spectroscopy II

Suppression And Inversion		
Water/fat	$\Delta\delta=3.5$ ppm	Water 4.7, fat 1.2 ppm.
Phase accrual	$\Delta\phi=2\pi\Delta\nu\tau$	Chemical shift phase.
Binomial spacing	$\tau\approx 1/(2\Delta\nu)$ or $1/(4\Delta\nu)$	Depends on target phase.
Inversion recovery	$M_z(TI)=M_0[1-2\exp(-TI/T_1)]$	After 180°.
Null	$T_{1\text{ null}} = T_1 \ln 2$	Water/fat suppression.

Localization		
Surface coil	$S(r)=B_1(r) \sin[\gamma B_1(r)t]$	Sensitivity + flip angle.
Adiabatic	$ d\theta/dt \ll \gamma B_{\text{eff}} $	Robust inversion/excitation.
PRESS/STEAM	Echo at TE; stimulated echo stores Mz during TM	Voxel selection.
Phase cycling	linear combinations isolate voxel term	Remove unwanted echoes.

CSI And High Field		
Shift artifact	$\Delta x=\Delta\nu_{CS}/(\gamma G)$	Chemical shift displacement.
Fractional shift	$\Delta x/\Delta x_{RF}=\Delta\nu_{CS}/BW_{RF}$	Use large RF BW.
CSI signal	$s(k_x,k_y,t)=\iiint \rho(x,y,\Delta\nu) \exp[i(k_x x+k_y y+2\pi\Delta\nu t)] \, dx dy d\Delta\nu$	2 spatial + spectral.
CSI recon	$S(x,y,\Delta\nu)=F^{-1}_t F^{-1}_{k_x} F^{-1}_{k_y}\{s\}$	3D transform.
CSI scan time	$T_{scan} = N_x N_y TR \cdot NSA$	Slow phase encoding.
High-field SNR	$U_{sig} \sim B_0^2 \cdot U_{noise} \sim B_0^4, \text{ SNR} \sim B_0$	Approximate.
SAR	$P \sim \sigma E^2 \sim \sigma \gamma^2 B_0^2 B_1^2$	High-field cost.